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ANRF AISEHack 2026

*Finale presentation /
Theme : India in the
Haze (PM2.5)*

Raj Shah, Saptarshi Misra,

Hiteshri Shastri | Team RuVision

ConvLSTM + UNet trained to forecast PM2.5 16h ahead from 10h lookback. Final score: 0.8636. Key innovations: simplified architecture (1-layer ConvLSTM, 2-level UNet, no CBAM), K=16 spatial clustering for loss weighting, and explicit episode-aware loss ($\alpha=1.0$, $\beta=3.0$, $\gamma=0.15$).

Your salient contributions

f1

Architecture: ConvLSTM + 2-level UNet (0.6M params)

f2

Training: 28 epochs, AdamW, OneCycleLR, K=16 cluster, loss weights

f3

Score: 0.8636 | +0.0233 over previous v6

Modelling - *Strategies*

01

Architecture

1-layer ConvLSTM (dim=48) + 2-level UNet (base=24).
No CBAM. 0.6M params.

AdamW + OneCycleLR, 28 epochs, batch=16. 4 months
WRF-Chem 2016.

01

02

Loss Function

$\alpha=1.0 \cdot \text{GlobalSMAPE} + \beta=3.0 \cdot \text{EpSMAPE} + \gamma=0.15 \cdot$
(1-EpCorr). Smooth SMAPE, pseudo-Huber $\tilde{\delta}=0.1$.

02

03

Clustering

KMeans K=16 on PM2.5 + lat/lon. Weights from mean,
episode freq, variance.

03

**Input (10h, 20ch) $\rightarrow$ ConvLSTM $\rightarrow$
2-level UNet $\rightarrow$ +Persistence $\rightarrow$
Forecast (16h)**

Strengths: Lightweight (0.6M), fast training (~ 15 min/epoch on T4), no CBAM overfitting, episode-aware loss.

Weaknesses: Single model may miss regional variation.
Regional ensemble attempted but did not improve.:

Key Results – *Why, what and how of the workings?*

S1

What is working?

ConvLSTM+UNet (v7): 1-layer LSTM dim=48, 2-level UNet base=24, no CBAM. Loss: $\alpha=1.0 \cdot \text{GlobalSMAPE} + \beta=3.0 \cdot \text{EpSMAPE} + \gamma=0.15 \cdot (1 - \text{EpCorr})$. K=16 spatial clusters.

S2

How is it working? (Quantitative evals)

v6→v7 gains: Arch simplification +0.014, K=16 clustering +0.005, explicit $\beta=3.0$ episode loss +0.021. Combined score: 0.8636. Val metrics: ng=0.86+, ne=0.87+, nc=0.85+

S3

Why is it working? (Reasoning and analysis)

Simpler arch reduces overfitting on ~2000 training samples. Explicit EpSMAPE term directly optimises the hardest competition sub-metric. K=16 clusters give finer spatial loss weighting (IGP vs coast vs NE India).

S4

0.6M params | Training: ~15 min on T4 GPU, 28 epochs total | Inference: ~2s per test batch (TTA: E-W flip). Phase 1→2 shift: moved from IGP binary mask to K=16 data-driven spatial clusters; replaced episode ramp with explicit β -EpSMAPE term; reduced model capacity 5M→0.6M params to prevent overfitting.

Key Results - *Why, what and how of the workings?*

01

Experiment Scores

v6 baseline: 0.8403 | Exp1-B (slim arch): 0.8542 |
Exp3-B (K=16): 0.8449 | Exp4-D ($\beta=3.0$): 0.8608 | v7
combined: 0.8636 | Regional: 0.8552
Regional ensemble: 0.8552. Exp2/5/6 showed
diminishing returns.

01

02

v7 Architecture

1-layer ConvLSTM + 2-level UNet + no CBAM = 0.6M
params vs v6's 5M. Simpler model generalises better on
~2000 training samples.

02

03

Phase 1 → Phase 2

Phase 1 used a generic loss function with experiments
on Huber, Focal and MSE. Phase 2 used score metric
defined as loss function and fine tuned the weights.

03

Leaderboard score: 0.8636.

**Score = $w_1 \cdot \text{NormGSMAPE} +$
 $w_2 \cdot \text{NormEpCorr} + w_3 \cdot \text{NormEpSMAPE}$**

Strengths: Fast convergence, lightweight, episode-aware
loss directly aligns training with eval metric.

Weaknesses: Regional ensemble did not improve
(0.8552). Could not experimented with lighter regional
models due to time constraint.

Details of Relevant Experimentation and Methodology

01

Ablations (sequential, each vs v6=0.8403)

Arch -0.014, K=16 +0.005, β =3.0 +0.021.

Exp2/5/6/regional: no gain.

v6 → 4 ablation experiments → v7 → regional → presentation.

01

02

References & Inspiration

ConvLSTM (Shi et al. 2015) for spatiotemporal forecasting. CBAM attention (Woo et al. 2018) — tested and found to overfit at this data scale. Pseudo-Huber smooth SMAPE for stable gradient near zero.

02

03

GenAI Tool Usage (Claude by Anthropic)

Claude used for: iterative notebook generation, loss function derivation from competition metric, and submission documentation. All model decisions validated empirically on leaderboard.

03